SOLAR EV CHARGING WITH COST MONITORING AND LOAD SWITCHING

A MINI-PROJECT REPORT

22UEE605 – MINI PROJECT II

Submitted by

HARISH R	**710723105016**
KISHOR S	**710723105027**
MITHUN R	**710723105032**
MUKILAN K	**710723105036**

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BONAFIDE CERTIFICATE

Certified that this 22UEE605 - Mini Project II report **“SOLAR EV CHARGING WITH COST MONITORING AND LOAD SWITCHING”** is the bonafide work of **“HARISH R (710723105016), KISHOR S (710723105027), MITHUN R (710723105032), MUKILAN K (710723105036)”** who carried out the project work under my supervision.

SIGNATURE	**SIGNATURE**
SUPERVISOR	**HEAD OF THE DEPARTMENT**
Mr. S.S. Karthikeyan	**Dr. K. Ramash Kumar**
Assistant Professor (SS)	Professor & Head
Department of EEE	Department of EEE
Dr. N.G.P. Institute of Technology	Dr. N.G.P. Institute of Technology
Coimbatore - 641048.	Coimbatore - 641048.

Submitted for the End-Semester Mini Project-II Viva- Voce examination held on

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Internal Examiner	**External Examiner**

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ABSTRACT

The rapid growth of Electric Vehicles (EVs) has created a significant demand for efficient, reliable, and cost-effective charging infrastructure. Conventional EV charging systems are primarily dependent on grid power, which leads to increased electricity costs and limited utilization of renewable energy sources. To address these challenges, this project presents the design and implementation of an ESP32-based Smart EV Charging System integrated with solar energy and intelligent power management. The proposed system utilizes solar photovoltaic panels as the primary energy source, supported by a battery storage system regulated through a PWM charge controller. An inverter is employed to convert DC power into AC, enabling compatibility with practical EV charging applications. The system also incorporates a 2-channel relay module controlled by the ESP32 microcontroller to enable automatic switching between EV charging and auxiliary loads, ensuring efficient energy distribution. The ESP32 continuously monitors voltage, current, charging duration, and system status using appropriate sensors. It provides real-time feedback through an LCD display, including battery percentage, charging time, and cost estimation. The system ensures safe operation through controlled charging, overvoltage protection, and efficient load management. The proposed solution offers a sustainable, scalable, and intelligent approach to EV charging, reducing dependency on conventional power sources and promoting the use of renewable energy.

Key words: ESP32 Microcontroller, Smart EV Charging, Solar Energy, PWM Charge Controller, Inverter, Relay Switching, Energy Management, IoT Monitoring**.**

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LIST OF ABBREVIATIONS

Abb	Abbreviation
EV	Electric Vehicle
ESP32	Espressif Systems 32-bit Microcontroller
IoT	Internet of Things
LCD	Liquid Crystal Display
AC	Alternating Current
DC	Direct Current
Wi-Fi	Wireless Fidelity
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit
PWM	Pulse Width Modulation
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

The increasing demand for sustainable transportation has significantly accelerated the adoption of Electric Vehicles (EVs) worldwide. As EV usage continues to expand, the need for efficient, intelligent, and affordable charging infrastructure becomes increasingly important. Conventional EV charging systems rely heavily on grid electricity, which leads to higher operational costs and increased carbon emissions due to dependence on non-renewable energy sources. To address these challenges, the integration of renewable energy—particularly solar energy—into EV charging systems has emerged as a promising solution. Solar energy is a clean, abundant, and sustainable power source that can reduce reliance on conventional grid power while lowering operational expenses. However, solar energy is inherently intermittent, as its availability depends on environmental conditions such as sunlight intensity and time of day. Therefore, an efficient energy management system is essential to ensure continuous and reliable EV charging. This project proposes a **Smart EV Charging System using the ESP32 microcontroller**, integrating solar energy, battery storage, and intelligent control mechanisms. The system utilizes a PWM charge controller to regulate solar power and safely charge the battery. An inverter is used to convert stored DC power into AC power for practical EV charging applications. Additionally, a 2-channel relay module enables automatic switching of loads, ensuring optimal utilization of available energy. The ESP32 acts as the central control unit, continuously monitoring parameters such as voltage, current, and charging status. It processes data in real time and controls system operations to achieve efficient energy management. A display unit (LCD) is incorporated to provide users with real-time information regarding system performance and charging conditions. By combining renewable energy with smart monitoring and control, this system aims to deliver a reliable, efficient, and environmentally friendly EV charging solution. It reduces dependency on conventional power sources and promotes sustainable energy utilization, making it suitable for residential and small-scale commercial applications.

1.1 Background and Motivation

Growing concerns about environmental pollution and climate change have driven the global transition toward green energy solutions. Electric Vehicles have become a key component of sustainable transportation; however, their overall environmental benefits depend heavily on the availability of clean charging infrastructure. Most existing EV charging stations rely on grid electricity, which often originates from non-renewable energy sources. This limits the potential environmental advantages of EV adoption. Integrating solar energy into EV charging systems offers a pathway toward truly sustainable mobility. However, the intermittent nature of solar energy poses challenges in maintaining consistent charging performance. This creates the need for a hybrid and intelligent energy management system capable of balancing multiple power sources while ensuring uninterrupted operation. This project is motivated by the need to develop such a system that enhances efficiency, reliability, and sustainability.

1.2 Wireless Communication and Control

The ESP32 microcontroller, equipped with built-in Wi-Fi and Bluetooth capabilities, plays a crucial role in enabling smart monitoring and control of the charging system. It supports real-time data acquisition and system management, making the system more intelligent and responsive. The system can be integrated with Internet of Things (IoT) platforms, allowing users to monitor and control the charging process remotely. Parameters such as battery status, charging progress, and energy consumption can be accessed from anywhere, improving convenience and transparency.

Furthermore, IoT integration enables future enhancements such as mobile application control, cloud-based data storage, and predictive maintenance. These features contribute to improved system efficiency, reliability, and user experience.

1.3 Hybrid Power Management

The proposed system incorporates a hybrid energy management approach to efficiently utilize solar energy while ensuring reliable operation through battery storage and controlled power distribution. A relay-based switching mechanism is used to manage power flow between different components. The system prioritizes solar energy usage whenever available and automatically switches to stored battery power or alternative sources when necessary. This ensures uninterrupted charging and optimal energy utilization. Efficient energy balancing not only improves system performance but also extends battery life. By minimizing energy wastage and optimizing resource allocation, the system becomes suitable for long-term and practical applications.

1.4 Scope of the Project

This project focuses on the design and implementation of a smart solar-based EV charging system using the ESP32 microcontroller. It aims to provide an efficient, cost-effective, and environmentally friendly charging solution with real-time monitoring and intelligent control features. The system serves as a prototype for scalable EV charging infrastructure that can be expanded for larger applications. It demonstrates the feasibility of integrating renewable energy with modern control technologies to support sustainable transportation.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The literature survey provides a detailed understanding of the recent developments and research trends in electric vehicle (EV) charging systems, particularly focusing on IoT-based smart charging, renewable energy integration, and intelligent energy management. With the rapid increase in EV adoption, there is a growing demand for efficient, reliable, and sustainable charging infrastructure. Traditional EV charging systems are mostly dependent on grid power and lack intelligent monitoring, resulting in higher operational costs, inefficient energy utilization, and limited scalability. Recent advancements in IoT technologies have enabled the development of smart EV charging systems that support real-time monitoring, remote control, and automated decision-making. Microcontrollers such as ESP32 play a crucial role in enabling wireless communication and integrating sensors for monitoring voltage, current, and battery status. These systems improve user convenience by providing real-time information such as charging progress, battery percentage, and cost estimation. In addition to IoT integration, renewable energy sources such as solar power are increasingly being incorporated into EV charging systems. Solar-powered charging stations reduce dependency on conventional electricity sources and promote sustainable energy usage. The integration of battery storage systems further enhances reliability by ensuring uninterrupted power supply during low solar availability. Energy management strategies are also being developed to efficiently distribute power between EV charging and other loads. Furthermore, advanced control techniques such as DC-DC converters and smart grid integration have been explored to improve system efficiency and stability. These technologies enable optimal energy conversion, load balancing, and peak load reduction. Communication protocols and cloud-based platforms facilitate seamless data transmission and enable remote monitoring and control of charging stations. Security and safety aspects, including overload protection, fault detection, and user authentication, are also essential components of modern EV charging systems. Overall, the literature highlights that combining IoT, renewable energy, and intelligent control techniques significantly enhances the performance, scalability, and sustainability of EV charging infrastructure.

[1] S. Dhavale, P. Patil, and R. Kulkarni, "Development of an IoT-Based Adaptive Charging System for Electric Vehicles," International Journal of Advanced Research in Engineering and Technology (IJARET), vol. 16, no. 2, pp. 112–118, Feb. 2025.

DESCRIPTION:

This paper presents an IoT-based adaptive charging system for electric vehicles that improves both efficiency and flexibility. The system uses sensors and a microcontroller to monitor battery level, voltage, and current in real time, while IoT integration allows users to remotely track and control the charging process through a connected platform. By optimizing energy usage and reducing operational costs, the system ensures better performance. It also dynamically adjusts charging parameters based on real-time conditions, providing enhanced convenience and reliability for users.

[2] J. Srinivas, M. Reddy, and K. Prasad, "IoT-Enabled Smart EV Charging System with Real-Time Monitoring and Cost Optimization," International Journal of Electrical and Electronics Engineering Research (IJEEER), vol. 15, no. 3, pp. 201–208, Mar. 2025.

DESCRIPTION:

This research introduces a smart EV charging system that incorporates IoT technology for real-time monitoring and cost optimization. It also calculates the total cost of charging based on energy usage, ensuring transparency for users. The system provides real-time updates through a mobile or web interface, enhancing user convenience. Additionally, the study emphasizes efficient energy management, reducing unnecessary power consumption and improving the overall performance of EV charging stations. The system enhances decision-making by providing accurate data insights for efficient charging management. It also supports sustainable energy usage by optimizing power distribution and reducing energy wastage.

[3] A. Kumar, S. Verma, and R. Singh, "Smart Grid Integrated EV Charging System with Renewable Energy Sources," IEEE Transactions on Smart Grid, vol. 15, no. 1, pp. 55–63, Jan. 2024.

DESCRIPTION:

This paper explores the integration of electric vehicle charging systems with smart grid technology and renewable energy sources such as solar power. The proposed system effectively balances energy between the grid and renewable sources, ensuring stable and efficient operation. The study highlights how smart grids can support large-scale EV adoption by optimizing energy distribution and reducing dependence on fossil fuels. This work plays a significant role in promoting sustainable and intelligent power systems. The system improves grid stability by intelligently managing load variations between renewable sources and EV demand. It also enhances reliability by ensuring continuous power availability through efficient energy coordination.

[4] R. Sharma and P. Gupta, “Design and Implementation of a Microcontroller-Based EV Charger with Protection Features,” International Journal of Power Electronics and Drive Systems (IJPEDS), vol. 14, no. 2, pp. 89–96, 2023.

DESCRIPTION:

This study focuses on the development of a microcontroller-based EV charging system equipped with multiple protection features to ensure safe operation. These features help prevent damage to both the battery and the charging system. The use of a microcontroller allows precise control of the charging process, improving reliability and performance. The research emphasizes the importance of safety in EV charging systems, making it suitable for both residential and industrial applications. The system ensures enhanced operational safety by incorporating advanced fault detection and protection mechanisms. It also improves charging efficiency through accurate control and monitoring of charging parameters.

[5] Y. Zhang, H. Liu, and X. Chen, “An Intelligent Solar Powered EV Charging System with Battery Storage,” Renewable Energy Journal, vol. 190, pp. 120–130, 2022.

Description:

This paper presents a solar-powered EV charging system integrated with battery storage to ensure uninterrupted operation. The system stores excess solar energy during peak sunlight hours and uses it when solar generation is low or unavailable. The study also highlights the environmental benefits of using renewable energy sources, such as reduced carbon emissions and lower operational costs. The proposed system is ideal for sustainable and eco-friendly EV charging infrastructure.

2.2 Literature Review Summary

Reference	**Title**	**Key Focus**	**Findings**	**Impact on Industrial Automation**
[1] S. Dhavale et al.	Development of an IoT-Based Adaptive Charging System for Electric Vehicles	IoT-based adaptive charging system with real-time monitoring of battery parameters	Dynamically adjusts charging rate, improves efficiency, and prevents overcharging	Enables intelligent and automated EV charging systems with improved energy utilization
[2] J. Srinivas et al.	IoT-Enabled Smart EV Charging System with Real-Time Monitoring and Cost Optimization	Real-time monitoring of energy consumption and cost calculation using IoT	Provides accurate billing, transparency, and optimized energy usage	Supports cost-effective and automated charging infrastructure
[3] A. Kumar et al.	Smart Grid Integrated EV Charging System with Renewable Energy Sources	Integration of renewable energy with smart grid for EV charging	Achieves load balancing and reduces dependency on conventional power sources	Promotes sustainable and automated power distribution systems
[4] R. Sharma et al.	Design and Implementation of a Microcontroller-Based EV Charger with Protection Features	Microcontroller-based charger with safety features like overvoltage and overcurrent protection	Ensures safe and reliable charging with fault detection mechanisms	Improves safety and reliability in industrial EV charging applications
[5] Y. Zhang et al.	An Intelligent Solar Powered EV Charging System with Battery Storage	Solar-based EV charging with battery backup system	Provides continuous charging and reduces grid dependency	Enhances renewable energy usage in automated charging stations

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

The existing electric vehicle (EV) charging systems are primarily based on conventional grid-powered infrastructure with limited intelligence and automation. These systems are widely used in residential, commercial, and public charging stations. However, they often lack advanced monitoring, control, and energy optimization features. Most traditional EV charging setups operate independently without integrating renewable energy sources or smart control mechanisms, leading to inefficient energy utilization and increased operational costs. In conventional systems, EV charging is directly connected to the electrical grid, and the charging process begins manually when the vehicle is plugged in. These systems do not provide real-time information about battery status, charging duration, or energy consumption. As a result, users are unable to monitor charging progress effectively. Moreover, the absence of automated control systems makes it difficult to manage multiple loads or optimize energy usage during peak hours. With the increasing demand for EVs, the limitations of existing systems have become more evident. The lack of scalability, high installation costs, and absence of intelligent energy management reduce the overall efficiency of charging infrastructure. Furthermore, traditional systems do not support integration with solar energy or battery storage, which are essential for sustainable and cost-effective operation. These challenges highlight the need for advanced EV charging systems that incorporate IoT, automation, and renewable energy technologies.

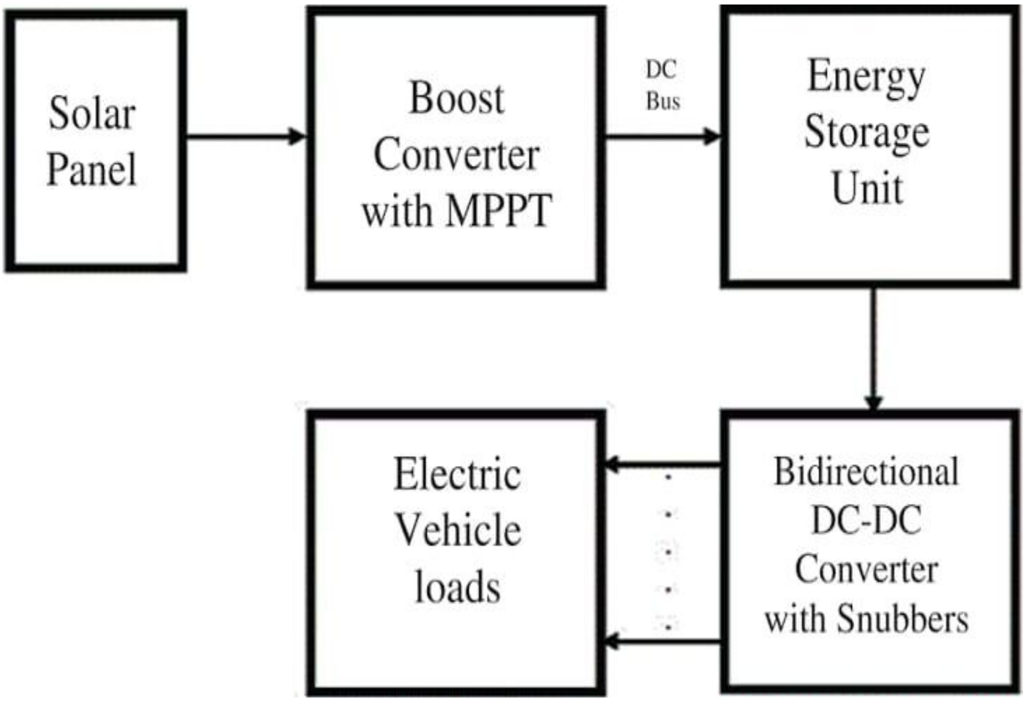

Fig. 3.1 Block Diagram of the Existing System

3.2 WORKING OF THE EXISTING SYSTEM

The working of a conventional EV charging system is relatively simple and follows a basic electrical connection model. When an electric vehicle is connected to the charging point, power is supplied directly from the grid to the vehicle battery through a charging unit. The charging unit typically includes basic protection circuits such as fuses and circuit breakers to ensure safe operation. In most existing systems, the charging process is manually controlled, and there is no intelligent monitoring of parameters such as voltage, current, or battery percentage. The system does not provide feedback to the user regarding charging status or estimated completion time. This lack of real-time monitoring reduces user convenience and system transparency. Additionally, traditional systems do not include automatic load management. During peak hours, multiple charging stations may draw excessive power from the grid, leading to increased electricity costs and potential overload conditions. There is also no mechanism to prioritize loads or switch between different power sources. Another limitation is the absence of communication between system components. Without IoT integration, the system cannot support remote monitoring or control. Maintenance is also reactive rather than proactive, as faults are detected only after failure occurs. This leads to increased downtime and higher maintenance costs.

3.3 DEMERITS OF THE EXISTING SYSTEM

Grid Dependency: Existing EV charging systems rely entirely on grid power, increasing electricity costs and limiting sustainability.

No Renewable Energy Integration: Most systems do not utilize solar or alternative energy sources, leading to higher carbon emissions.

Lack of Real-Time Monitoring: Users cannot track battery percentage, charging time, or energy consumption during charging.

No Cost Calculation: Traditional systems do not provide automatic billing or cost estimation, reducing transparency.

Manual Operation: Charging process requires manual control without automation or intelligent decision-making.

No Load Management: Inability to manage multiple loads leads to inefficient energy distribution and possible overload.

Limited Scalability: Difficult to expand or upgrade existing systems due to rigid infrastructure.

Higher Operational Cost: Inefficient energy usage results in increased electricity bills.

No Remote Monitoring: Users and operators cannot monitor or control the system remotely.

Lack of Backup Support: No battery storage or backup system leads to interruption during power failure.

3.4 SUMMARY

The existing EV charging systems are limited by their dependence on grid power, lack of intelligent monitoring, and absence of automation. These systems do not support real-time data tracking, efficient energy management. The inability to integrate renewable energy sources further reduces their sustainability and increases operational costs. Additionally, the lack of load management and remote monitoring capabilities affects system performance and user convenience. These limitations highlight the need for a smart EV charging system that incorporates IoT technology, renewable energy integration.

CHAPTER 4

PROPOSED SYSTEM

4.1 INTRODUCTION

The proposed system is a smart solar-based EV charging station designed to improve charging efficiency and optimize energy utilization. The system uses an ESP32 microcontroller for intelligent control and monitoring. It integrates renewable energy through solar panels and manages power using a PWM controller and inverter system. The PWM charge controller regulates the voltage from the solar panel and ensures efficient battery charging. The inverter converts DC power into AC power, making it suitable for real-world EV charging applications. A 2-channel relay module is used for automatic load switching between EV charging and other loads. The system detects the EV battery using a voltage sensor and initiates charging automatically. It continuously monitors voltage, current, charging time, and cost. This makes the system more efficient, reliable, and suitable for smart applications.

4.2 Block Diagram

The solar panel generates DC power, which is regulated by the PWM charge controller. The stored energy is then passed through an inverter to convert it into AC. The ESP32 monitors system parameters and controls the 2-channel relay. One relay channel supplies power to the EV charger, while the other manages additional loads. The LCD displays real-time system data.

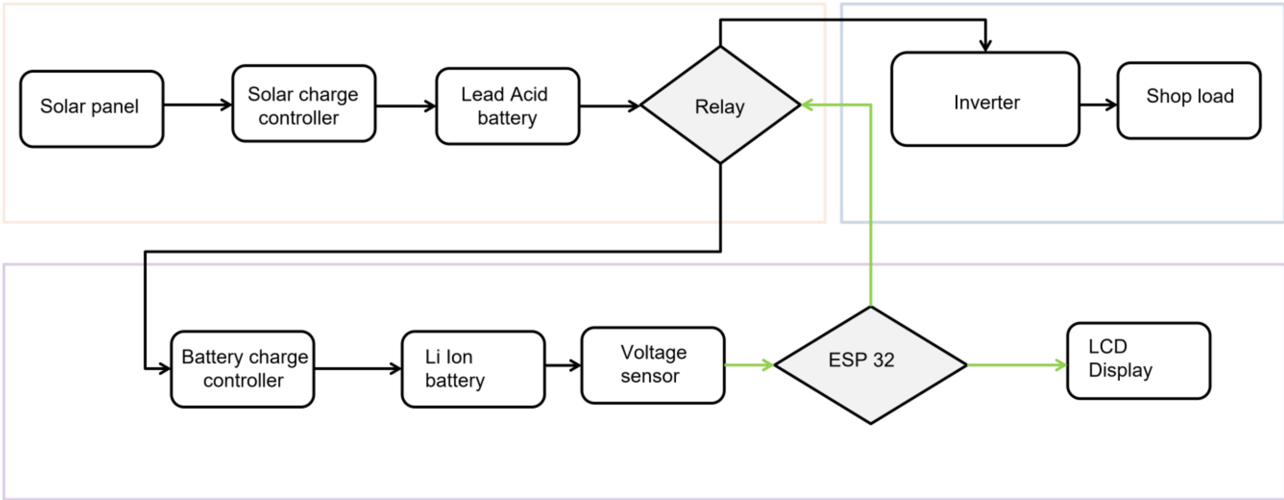

Fig. 4.1 Block Diagram of the Proposed System

4.3 Working Methodology

1. Power Generation

- Solar panel generates DC power
- PWM controller regulates voltage

2. Energy Conversion

- Inverter converts DC to AC power
- Makes system suitable for EV charging

3. Battery Detection

- Voltage sensor detects EV connection
- ESP32 initiates charging

4. Charging Control

- 2-channel relay switches ON EV charging
- Controls load distribution

5. Monitoring

- Sensors measure voltage and current
- Data processed in real-time

6. Cost and Time Calculation

- Charging duration tracked
- Cost calculated based on power usage

7. Display Output

- LCD shows:
    - Battery percentage
    - Charging time
    - Cost

4.4 Design Considerations

Components Main:

ESP32 Microcontroller
Solar Panel
PWM Charge Controller
Inverter
Voltage Sensor
Current Sensor
2-Channel Relay Module
LCD Display
EV Battery

System Features:

1.ESP32 Controller:

Acts as the main control unit.

Processes sensor data.

Controls relay switching.

2.PWM Charge Controller:

Regulates solar panel output.

Prevents overcharging.

Improves charging efficiency.

3.Inverter:

Converts DC power into AC.

Enables real-world EV charging compatibility.

4.Voltage Sensor:

Detects battery connection.

Measures voltage levels.

6. 2-Channel Relay Module:

One channel controls EV charging.

Another channel controls auxiliary load (shop/home load).

Provides automatic switching.

7.LCD Display:

Displays battery percentage, time, and cost.

4.5 COMPONENTS

1. ESP32 Microcontroller

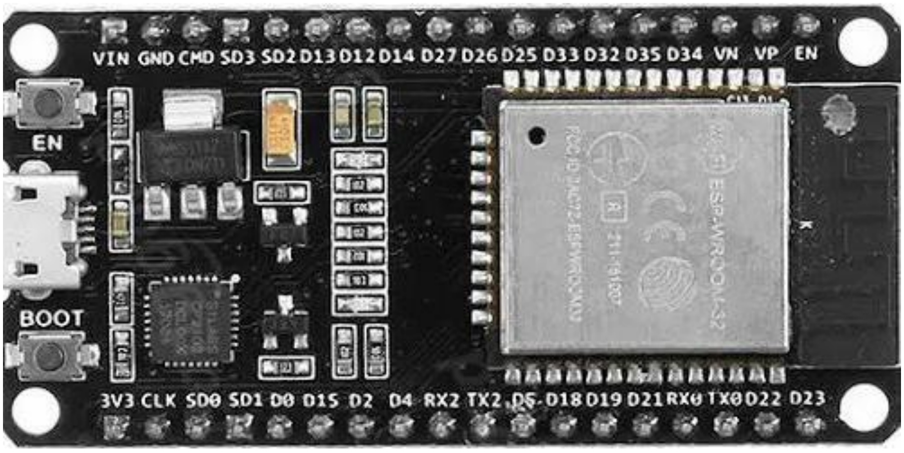

Fig. 4.2 ESP32 Controller

The ESP32 microcontroller serves as the central processing and communication unit of the system. It is responsible for monitoring sensor inputs, executing the source switching logic communicating with the Blynk IoT platform, driving the LCD display and managing all timing-critical operations. Its integrated Wi-Fi capability eliminates the need for an external wireless module, significantly reducing system complexity and cost.

Table 2: ESP32 Microcontroller Specifications

Feature	Specification
Processor	Xtensa Dual-Core 32-bit LX6 CPU, 240 MHz
Wireless Connectivity	Wi-Fi (802.11 b/g/n) and Bluetooth v4.2 (BLE + Classic)
Memory	520 KB SRAM, up to 16 MB external Flash
GPIO & Interfaces	34 GPIO pins, 18 ADC channels, 3 UARTs, 4 SPI, 2 I2C
Operating Voltage	3.3 V (logic), 5 V tolerant inputs
Power Efficiency	Deep sleep mode (~5 µA consumption)

2. Relay Module

Fig. 4.3 Relay Module

A dual-channel relay module is employed to switch between the solar and grid power inputs. Each relay channel is driven by a digital output pin of the ESP32 through an NPN transistor driver circuit, providing electrical isolation between the logic circuitry and the high-voltage switching contacts. Flyback diodes protect the driver transistors from voltage spikes generated during relay coil de- energization.

Table 3: Relay Module Specifications

Feature	Specification
Coil Voltage	5 V DC
Contact Rating	10 A at 250 V AC / 30 V DC
Control Signal	Active-low (triggered by logic low from ESP32 via transistor)
Channels	2 (one per power source)
Protection	Flyback diode integrated on module

3. 16x2 I2C LCD Display

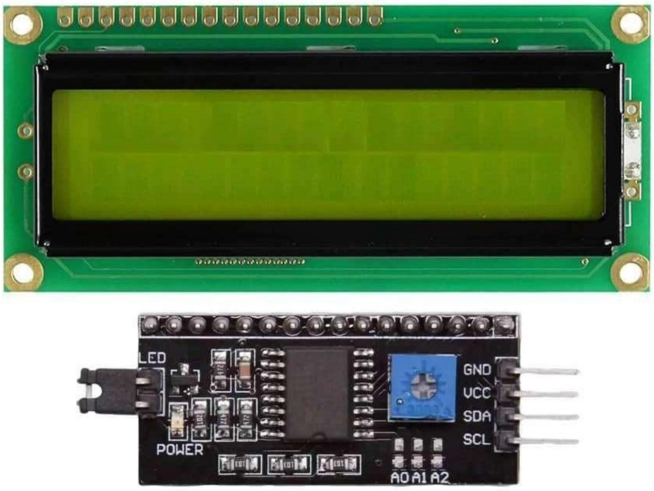

Fig. 4.4 16x2 I2C LCD Display

The 16x2 character LCD display with I2C interface module is used to present real-time system status to the user. The I2C interface reduces the number of GPIO pins required to drive the display from eight (in parallel mode) to two (SDA and SCL), conserving valuable ESP32 I/O resources. The LCD is configured to display operating mode, grid status, battery voltage and charging percentage across its two rows.

Table 4: 16x2 I2C LCD Specifications

Feature	Specification
Display Type	16 characters × 2 lines, character LCD
Interface	I2C (via PCF8574 I/O expander)
Operating Voltage	5 V DC
I2C Address	0x27 (default, configurable)
Backlight	LED (controllable via I2C)

4. INVERTER

Fig. 4.5 Inverter

An inverter is an electrical device that converts direct current (DC) into alternating current (AC). It is commonly used in systems where the power source, such as a battery or solar panel, provides DC power but the appliances require AC power to operate. Inverters are widely used in homes, vehicles, and renewable energy systems to run devices like lights, fans, and chargers. They play an important role in providing backup power during outages and enabling the use of portable or stored energy sources.

Table 5: Inverter Specifications

Feature	**Specification**
Input Voltage	12V DC
Output Signal	Modified Sine Wave
Isolation	Non-isolated
Sensitivity	Medium load sensitivity
Operating Voltage	10V – 15V DC

5. 25V DC Voltage Sensor

Fig. 4.6 25V DC Voltage Sensor

A DC voltage sensor module (0–25 V range) is used to monitor the battery voltage. It employs a resistive voltage divider to scale down the input voltage to a level compatible with the ESP32 ADC (0–3.3 V). This allows real-time monitoring of battery condition and supports accurate estimation of the state of charge.

Table 6: DC Voltage Sensor Specifications

Feature	Specification
Input Voltage	0 – 25 V DC
Output Voltage	0 – 3.3 V (scaled)
Measurement Type	Analog
Accuracy	±1% (approx.)
Interface	ADC (ESP32 compatible)

6. **Buck Converter (LM2596)**

Fig. 4.7 Buck Converter (LM2596)

The LM2596 buck converter is used to step down higher DC voltages to a stable lower voltage required by system components such as the ESP32 and sensors. It ensures efficient power conversion with minimal heat dissipation, improving overall system reliability.

Table 7: LM2596 Buck Converter Specifications

Feature	Specification
Input Voltage	4 – 40 V DC
Output Voltage	1.25 – 37 V DC (adjustable)
Output Current	Up to 3 A
Efficiency	Up to 92%
Switching Frequency	150 kHz

7. **Lithium-Ion Battery (3.7 V, 1200 mAh)**

Fig. 4.8 Lithium-Ion Battery (3.7 V, 1200 mAh)

A 3.7 V lithium-ion battery is used as a portable energy storage element for low-power backup applications. It offers high energy density, lightweight design, and good rechargeability, making it suitable for IoT-based systems.

Table 8: Lithium-Ion Battery Specifications

Feature	Specification
Nominal Voltage	3.7 V
Capacity	1200 mAh
Chemistry	Li-ion
Rechargeable	Yes
Application	Backup power

8. Lead Acid Battery (12 V, 1.3 Ah)

Fig. 4.9 Lead Acid Battery (12 V, 1.3 Ah)

A 12 V sealed lead-acid battery is used for primary energy storage in the system. It provides stable output and is widely used in solar applications due to its reliability and cost-effectiveness.

Table 9: Lead Acid Battery Specifications

Feature	Specification
Nominal Voltage	12 V
Capacity	1.3 Ah
Type	Li-ion
Rechargeable	Yes
Application	Solar energy storage

9. PWM Solar Charge Controller (30A)

Fig. 4.10 PWM Solar Charge Controller (30A)

A 30A PWM charge controller regulates the charging of the battery from the solar panel. It prevents overcharging, deep discharge and improves battery life by maintaining proper charging conditions.

Table 10: PWM Controller Specifications

Feature	Specification
Rated Current	30 A
System Voltage	12 V / 24 V
Charging Method	PWM
Protection	Overcharge, over-discharge
Efficiency	Moderate

10. Solar Panel (20W)

Fig. 4.11 Solar Panel (5W)

A 20W solar panel is used to convert solar energy into electrical energy. It serves as the primary renewable energy source for the system.

Table 11: Solar Panel Specifications

Feature	Specification
Power Rating	5 W
Output Voltage	~18 V (open circuit)
Output Current	~0.28 A
Type	Polycrystalline/Monocrystalline
Application	Renewable energy generation

4.6 MERITS AND DEMERITS OF THE PROPOSED SOLUTION

Merits:

- **Renewable Energy Utilization:** Uses solar power, reducing dependence on conventional electricity and promoting eco-friendly charging.
- **Automatic Load Control:** The 2-channel relay enables intelligent switching between EV charging and auxiliary loads.
- **Efficient Battery Charging:** PWM charge controller ensures regulated charging and prevents overcharging of the battery.
- **Real-Time Monitoring:** ESP32 continuously monitors voltage, current, charging time, and system performance.
- **Cost Effective in Long Run:** Reduces electricity bills by utilizing solar energy efficiently.
- **User-Friendly Interface:** LCD display provides clear information about charging status, cost, and battery percentage.

Demerits:

- **High Initial Cost:** Installation of solar panels, inverter, and controller increases initial investment.
- **Dependence on Weather Conditions:** Solar power generation is affected by sunlight availability.
- **Energy Conversion Losses:** Inverter and PWM controller introduce power losses, reducing overall efficiency.
- **Limited Charging Speed:** Charging rate may be slower compared to grid-based fast charging systems.
- **Maintenance Requirement:** Solar panels and battery systems require periodic maintenance.
- **Less Efficient:** PWM are less efficient compared to MPPT controllers.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

The results and discussion section presents a detailed evaluation of the performance of the proposed smart EV charging station. The system is tested under different operating conditions to analyze its efficiency, reliability, and real-time performance. The experimental results demonstrate the effectiveness of the ESP32-based control system in managing EV charging operations, including battery detection, charging control, monitoring, and cost calculation. The system successfully detects battery connection using a voltage sensor and initiates charging automatically through relay control. Real-time monitoring of voltage and current enables accurate estimation of battery percentage and energy consumption. The integration of sensors and microcontroller ensures stable operation with minimal delay.

The LCD display provides clear and continuous feedback to the user, showing charging status, time, and cost. The system also demonstrates efficient load management by switching between EV charging and other loads. Safety features such as controlled power delivery and proper relay operation prevent system failures. Overall, the results confirm that the proposed system improves efficiency, reduces manual intervention, and enhances user convenience compared to traditional EV charging methods.

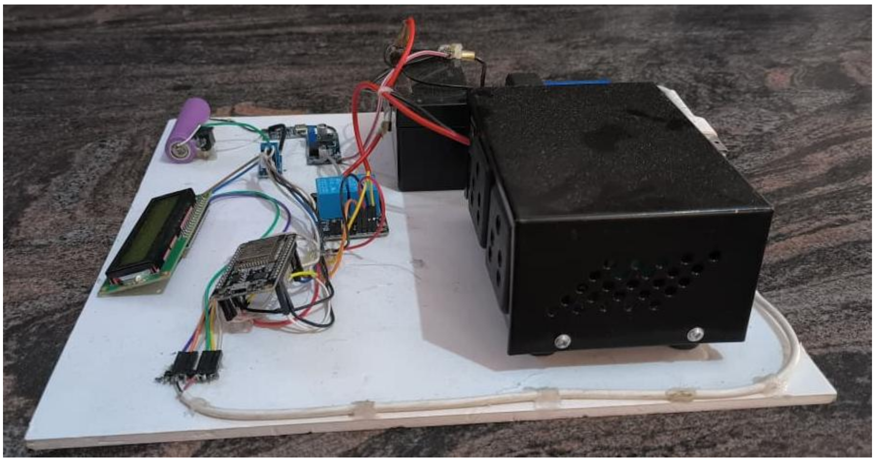

Fig. 5.1 Side View Of The Proposed System

5.2 CONTROL MODULE

The control module is the core unit of the system, responsible for processing inputs and controlling the charging process. It consists of an ESP32 microcontroller, voltage sensor, current sensor, relay module, and display unit. The ESP32 receives input signals from the sensors and processes them in real time. Based on the detected battery voltage, the controller decides whether to start or stop the charging process. The relay module is controlled by the ESP32 to switch the power supply to the EV battery.

The LCD display provides continuous updates regarding:

- Battery percentage
- Charging time

- Charging cost
- System status (Charging / Idle)

The control module operates efficiently with low latency and ensures smooth system performance. The use of wireless-enabled ESP32 allows future expansion for remote monitoring and IoT integration.

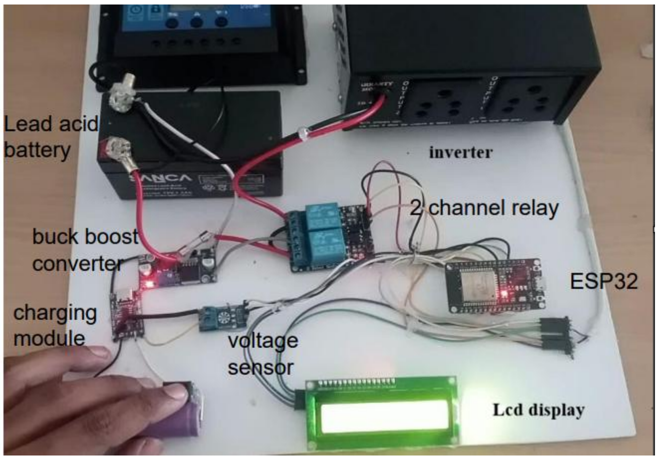

Fig. 5.2 Front View of the Proposed System

5.3 CHARGING MECHANISM

The charging mechanism ensures efficient and safe transfer of electrical energy to the EV battery.

1. Battery Detection and Initialization

The voltage sensor detects the presence of an EV battery. Once detected, the ESP32 activates the relay to begin charging. This automatic process eliminates the need for manual intervention.

2. Charging Operation

During charging:

- Voltage and current are continuously monitored
- Power is supplied in a controlled manner
- Relay ensures safe switching

The system maintains stable charging conditions to prevent overloading or fluctuations.

3. Monitoring and Feedback

The ESP32 calculates battery percentage based on voltage levels. Charging time is tracked using an internal timer. Energy consumption is calculated using voltage and current readings.

4. Cost Calculation

The charging cost is calculated using the formula:

- Energy consumed (kWh) = Voltage × Current × Time
- Cost = Energy × Unit price

The calculated cost is displayed on the LCD in real time, providing transparency to the user.

5.4 SYSTEM PERFORMANCE ANALYSIS

The performance of the system is evaluated based on several parameters:

1. Efficiency

The system efficiently manages charging operations with minimal power loss. The integration of sensors ensures accurate measurement and control.

2. Accuracy

Voltage and current readings provide reliable data for battery percentage and cost calculation. Minor variations may occur due to sensor tolerances.

3. Response Time

The system responds quickly to battery connection and disconnection. Relay switching occurs with negligible delay.

4. Reliability

The system operates consistently under different conditions. Proper wiring and stable power supply ensure uninterrupted operation.

5. Safety

The use of relay control prevents overloading. Controlled power delivery protects both the system and the EV battery.

5.5 FUNCTIONALITY AND AUTOMATION

The proposed system demonstrates a high level of automation:

- Automatically detects EV battery connection
- Starts and stops charging without manual control
- Continuously monitors charging parameters
- Displays real-time data to the user

- Calculates cost automatically
- Manages load switching efficiently

The automation reduces human effort and improves system efficiency. The system can be used in applications such as charging stations, campuses, and commercial buildings.

5.6 FUTURE ENHANCEMENTS AND APPLICATIONS

The system can be further enhanced by integrating advanced technologies:

- **IoT Integration:** Remote monitoring using mobile applications
- **Solar Integration:** Use of renewable energy for charging
- **Smart Billing:** Prepaid or RFID-based billing systems
- **Cloud Storage:** Data logging and analysis
- **AI-Based Optimization:** Predictive energy management
- **Fast Charging Support:** Improved charging speed

Applications:

- EV charging stations
- Colleges and campuses
- Office buildings
- Residential charging systems
- Smart city infrastructure

5.7 SUMMARY

- The results and discussions confirm that the proposed smart EV charging station operates efficiently and reliably. The system successfully integrates ESP32, sensors, and relay modules to provide automated charging, real-time monitoring, and cost calculation.

- The experimental results demonstrate improved performance compared to conventional charging systems. The system reduces manual effort, enhances energy management, and provides a user-friendly interface. The inclusion of automation and intelligent control makes it suitable for modern EV infrastructure.

- Overall, the proposed system offers a cost-effective, scalable, and efficient solution for EV charging applications. Future enhancements such as IoT and renewable energy integration can further improve system performance and sustainability.

CHAPTER 6
CONCLUSION

The proposed smart EV charging station system successfully demonstrates an efficient and automated solution for electric vehicle charging using modern embedded and sensing technologies. The system is designed using an ESP32 microcontroller, along with voltage and current sensors, relay modules, and a display unit, to provide real-time monitoring and intelligent control of the charging process. The system effectively detects the presence of an EV battery and initiates charging automatically. It continuously monitors important parameters such as voltage, current, battery percentage, charging time, and energy consumption. Based on these parameters, the system accurately calculates the charging cost, providing transparency and convenience to the user. One of the major achievements of this project is the implementation of automated load switching, which allows efficient utilization of available power between EV charging and other loads.

This feature makes the system suitable for small-scale applications such as shops, homes, and local charging stations. The use of sensors ensures reliable operation, while the relay-based control mechanism enhances safety by preventing overloading and improper connections. The system also proves to be cost-effective, easy to implement, and scalable for future improvements. Compared to traditional EV charging systems, this solution reduces manual effort, improves efficiency, and supports better energy management. The integration of real-time monitoring and automatic cost calculation enhances user experience and system usability.

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APPENDIX

CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include "BluetoothSerial.h"

LiquidCrystal_I2C lcd(0x27, 16, 2);
BluetoothSerial SerialBT;

// Pins
#define VOLT_PIN 34
#define RELAY_SHOP   26   // IN1 (Shop Load)
#define RELAY_CHARGE 27   // IN2 (Charging)

// Calibration
float voltageFactor = 11.0;

// Battery limits
float minBatteryVolt = 3.0;
float maxBatteryVolt = 4.2;
float detectThreshold = 2.5;

// Cost settings
float pricePerPercent = 0.50;

bool vehiclePresent = false;
bool prevVehiclePresent = false;
bool sessionActive = false;
bool batteryFull = false;

float initialPercent = 0;
float batteryPercent = 0;
```

```
float chargingPercent = 0;
float finalCost = 0;

// New Variables
float totalCost = 0;
int totalSessions = 0;
String systemStatus = "GRID";

int anim = 0;

void setup()
{
 Serial.begin(115200);

 pinMode(RELAY_CHARGE, OUTPUT);
 pinMode(RELAY_SHOP, OUTPUT);

 lcd.init();
 lcd.backlight();

 analogReadResolution(12);

 // Bluetooth Start
 SerialBT.begin("EV_CHARGER");

 // Default → Shop load ON
 digitalWrite(RELAY_CHARGE, LOW);
 digitalWrite(RELAY_SHOP, HIGH);
}

void loop()
{
 // ===== READ VOLTAGE =====
 int adc = analogRead(VOLT_PIN);
```

```
float sensedVolt = (adc / 4095.0) * 3.3;
float batteryVolt = sensedVolt * voltageFactor;

// ===== BATTERY % =====
batteryPercent =
    ((batteryVolt - minBatteryVolt) /
     (maxBatteryVolt - minBatteryVolt)) * 100.0;

batteryPercent = constrain(batteryPercent, 0, 100);

// ===== VEHICLE DETECT =====
if (batteryVolt > detectThreshold)
  vehiclePresent = true;
else
  vehiclePresent = false;

// ==============================================
// BATTERY CONNECTED
// ==============================================
if (vehiclePresent && !prevVehiclePresent)
{
  initialPercent = batteryPercent;
  sessionActive = true;
  batteryFull = false;

  systemStatus = "EV CHARGING";

  digitalWrite(RELAY_CHARGE, HIGH);
  digitalWrite(RELAY_SHOP, LOW);
}
// ==============================================
//  BATTERY FULL
// ==============================================
if (batteryPercent >= 100 && sessionActive)
```

```
{
  batteryFull = true;

  chargingPercent = batteryPercent - initialPercent;
  finalCost = chargingPercent * pricePerPercent;

  totalCost += finalCost;
  totalSessions++;

  systemStatus = "GRID";

  digitalWrite(RELAY_CHARGE, LOW);
  digitalWrite(RELAY_SHOP, HIGH);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Battery Full");

  lcd.setCursor(0, 1);
  lcd.print("Cost Rs:");
  lcd.print(finalCost, 2);

  delay(4000);

  sessionActive = false;
}

// ==========================================
//  BATTERY REMOVED
// ==========================================
if (!vehiclePresent && prevVehiclePresent && sessionActive)
{
  chargingPercent = batteryPercent - initialPercent;
  if (chargingPercent < 0) chargingPercent = 0;
```

```
  finalCost = chargingPercent * pricePerPercent;

  totalCost += finalCost;
  totalSessions++;

  systemStatus = "GRID";

  digitalWrite(RELAY_CHARGE, LOW);
  digitalWrite(RELAY_SHOP, HIGH);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Charged:");
  lcd.print((int)chargingPercent);
  lcd.print("%");

  lcd.setCursor(0, 1);
  lcd.print("Cost Rs:");
  lcd.print(finalCost, 2);

  delay(5000);

  sessionActive = false;
}

// =============================================
// NORMAL DISPLAY
// =============================================
lcd.clear();

// WAITING
if (!vehiclePresent && !sessionActive)
{
```

```
  systemStatus = "GRID";

  digitalWrite(RELAY_CHARGE, LOW);
  digitalWrite(RELAY_SHOP, HIGH);

  lcd.setCursor(0, 0);
  lcd.print("EV CHARGING");

  lcd.setCursor(0, 1);
  lcd.print("Waiting...");
}

// CHARGING
else if (vehiclePresent && !batteryFull)
{
  chargingPercent = batteryPercent - initialPercent;
  if (chargingPercent < 0) chargingPercent = 0;

  lcd.setCursor(0, 0);
  lcd.print("Battery:");
  lcd.print((int)batteryPercent);
  lcd.print("%");

  lcd.setCursor(0, 1);
  lcd.print("Charging");

  if (anim == 0) lcd.print(".");
  if (anim == 1) lcd.print("..");
  if (anim == 2) lcd.print("...");
  if (anim == 3) lcd.print("....");

  anim++;
  if (anim > 3) anim = 0;
}
```

```
  prevVehiclePresent = vehiclePresent;

  // ==============================================
  // SEND DATA TO PHONE
  // ==============================================

  SerialBT.println("------EV CHARGER------");

  SerialBT.print("Total EV Charged: ");
  SerialBT.println(totalSessions);

  SerialBT.print("Total Cost: Rs ");
  SerialBT.println(totalCost, 2);

  SerialBT.print("Status: ");
  SerialBT.println(systemStatus);

  SerialBT.print("Battery: ");
  SerialBT.print((int)batteryPercent);
  SerialBT.println("%");

  SerialBT.print("Charging: ");
  SerialBT.print((int)chargingPercent);
  SerialBT.println("%");

  SerialBT.print("Cost: Rs ");
  SerialBT.println(finalCost, 2);
  SerialBT.println("----------------------");

  delay(1000);
}
```